

Prolog Program for a Database with Name and DOB (AI)

Aim

To write a Prolog program to create a simple database of names and dates of birth, and to query information from it.

Objectives

- To represent facts in Prolog using a knowledge base.
- To retrieve all facts using backtracking with fail-driven loops.
- To query specific information by unifying variables with facts.
- To display formatted output using format/2.

Algorithm

- 1 Start.
- 2 Define facts of the form dob(Name, date(Year, Month, Day)) for each person.
- 3 Define a rule show_all to print every record using backtracking (fail loop).
- 4 Call show_all to list all name-DOB records in the database.
- 5 Query the database for a specific person's date of birth.
- 6 Unify the query variable with the matching fact.
- 7 Display the retrieved date of birth.
- 8 Stop.

Source Code (Prolog)

```
% Prolog program - Database with Name and Date of Birth
dob('Arun', date(2003, 5, 14)).
dob('Divya', date(2002, 11, 23)).
dob('Karthik', date(2004, 2, 9)).
dob('Meena', date(2003, 7, 30)).
% Rule to display all records
show_all :-
    dob(Name, date(Y, M, D)),
    format("~w -> ~w-~w-~w~n", [Name, D, M, Y]),
    fail.
show_all.
:- write('Name-DOB Database:'), nl,
   show_all,
   nl,
   dob('Karthik', DOB),
   format("Query: dob('Karthik', DOB) -> DOB = ~w~n", [DOB]).
```

Output

```
Name-DOB Database:
Arun -> 14-5-2003
Divya -> 23-11-2002
Karthik -> 9-2-2004
```

```
Meena -> 30-7-2003  
Query: dob('Karthik', DOB) -> DOB = date(2004,2,9)
```

Result

The Prolog program successfully created and queried a Name-DOB database, displaying all records and correctly retrieving Karthik's date of birth as date(2004,2,9).